Lab. 3

Ejercicio 9

$$v = \frac{g \times m}{c}\left(1 - e^{-\left(\frac{c \times t}{m}\right)}\right)$$

$g = 9.8\ m/s^2$
$c = 15\ kg/s$
$v = 35\ m/s$
$t = 9\ s$

$$f(m) = \frac{g \cdot m}{c}\left(1 - e^{-\left(\frac{c \cdot t}{m}\right)}\right) - v$$

Formula Falsa Posición:

$$x_i = \frac{a \cdot f(b) - b \cdot f(a)}{f(b) - f(a)}$$

Si: $f(a) \cdot f(x_i) < 0$, $b = x_{i_2}$

Si: $f(b) \cdot f(x_i) < 0$, $a = x_i$

- Iteración 1:

Calculando $a = 40$

$$f(40) = \frac{(9.8\ m/s^2)(40\ kg)}{15\ kg/s}\left(1 - e^{-\frac{(15\ kg/s)(9\ s)}{40\ kg}}\right) - 35\ m/s$$

$$f(40) = -9.760900159\ m/s$$

Calculando $b = 100$

$$f(100) = \frac{(9.8\ m/s^2)(100\ kg)}{15\ kg/s}\left(1 - e^{-\frac{(15\ kg/s)(9\ s)}{100\ kg}}\right) - 35\ m/s$$

$$f(100) = 13.39630297\ m/s$$

Calculando aproximación

$$x_i = \frac{40(13.39630297) - 100(-9.760900159)}{13.39630297 - (-9.760900159)}$$

$$x_i = 65.29036025$$

$$f(65.29036025) = \frac{(9.8\ m/s^2)(65.29036025)}{15\ kg/s}\left(1 - e^{-\frac{(15\ kg/s)(9\ s)}{65.29036025\ kg}}\right) - 35\ m/s$$

$$f(65.29036025) = 2.261274636\ m/s$$

$$f(40) \cdot f(65.29036025) < 0 \longrightarrow (-9.760900159)(2.261274636) = -22.07207595$$

$[a, x_i]$

- Iteración 2

Valor con $a = 40 \longrightarrow f(40) = -9.760900159\ m/s$

Valor con $b = 65.29036025 \longrightarrow f(65.29036025) = 2.261274636\ m/s$

Calculando aproximación

$$x_{i_2} = \frac{40(2.261274636) - 65.29036025(-9.760900159)}{2.261274636 - (-9.760900159)}$$

$$x_{i_2} = 60.53344634$$

$f(x_{i_2})$

$$f(60.53344634) = \frac{(9.8\ m/s^2)(60.53344634)}{15\ kg/s}\left(1 - e^{-\frac{(15\ kg/s)(9\ s)}{60.53344634\ kg}}\right) - 35\ m/s$$

$$f(60.53344634) = 0.296659778$$

$f(a) \cdot f(x_{i_2}) < 0$

$$f(40) \cdot f(60.53344634) < 0 \longrightarrow (-9.760900159)(0.296659778) = -2.895666475$$

$[a, x_{i_2}]$

- Iteración 3

Valor con $a = 40 \longrightarrow f(40) = -9.760900159$

Valor con $b = 60.53344634 \longrightarrow f(60.53344634) = 0.296659778$

Calculando aproximación

$$x_{i_3} = \frac{40(0.296659778) - 60.53344634(-9.760900159)}{0.296659778 - (-9.760900159)}$$

$$x_{i_3} = 59.92778774$$

$f(x_{i_3})$

$$f(59.92778774) = \frac{(9.8\ m/s^2)(59.92778774)}{15\ kg/s}\left(1 - e^{-\frac{(15\ kg/s)(9\ s)}{59.92778774\ kg}}\right) - 35\ m/s$$

$$f(59.92778774) = 0.03731750404$$

$f(a) \cdot f(x_{i_3}) < 0$

$$f(40) \cdot f(59.92778774) < 0 \longrightarrow (-9.760900159)(0.03731750404) = -0.3642524311$$

$[a, x_{i_3}]$

- Iteración 4

Valor con $a = 40 \longrightarrow f(40) = -9.760900159$

Valor con $b = 59.92778774 \longrightarrow f(59.92778774) = 0.03731750404$

Calculando aproximación

$$x_{i_4} = \frac{40(0.03731750404) - 59.92778774(-9.760900159)}{0.03731750404 - (-9.760900159)}$$

$$x_{i_4} = 59.85189074$$

$f(x_{i_4})$

$$f(59.85189074) = \frac{(9.8\ m/s^2)(59.85189074)}{15\ kg/s}\left(1 - e^{-\frac{(15\ kg/s)(9\ s)}{59.85189074\ kg}}\right) - 35\ m/s$$

$$f(59.85189074) = 0.004668425$$

$f(a) \cdot f(x_{i_4}) < 0$

$$f(40) \cdot f(59.85189074) < 0 \longrightarrow (-9.760900159)(0.004668425) = -0.045568030$$

$[a, x_{i_4}]$

- Iteración 5

Valor con $a = 40 \longrightarrow f(40) = -9.760900159$

Valor con $b = 59.85189074 \longrightarrow f(59.85189074) = 0.004668425$

Calculando aproximación

$$x_{i_5} = \frac{40(0.004668425) - 59.85189074(-9.760900159)}{0.004668425 - (-9.760900159)}$$

$$x_{i_5} = 59.84240055$$

$f(x_{i_5})$

$$f(59.84240055) = \frac{(9.8\ m/s^2)(59.84240055)}{15\ kg/s}\left(1 - e^{-\frac{(15\ kg/s)(9\ s)}{59.84240055\ kg}}\right) - 35\ m/s$$

$$f(59.84240055) = 0.000583613$$

$f(a) \cdot f(x_{i_5}) < 0$

$$f(40) \cdot f(59.85189074) < 0 \longrightarrow (-9.760900159)(0.000583613) = -0.005696588$$

$[a, x_{i_5}]$